
Algorithm 1: COMPUTE_DELTA_R(W, u, R, σ_1, W_0)

Input: RDB2RDF view $W = (V, S, M)$; update $u = (D, I)$ over relation R ; post-update database state σ_1 ; pre-update RDF state W_0 .

Output: Changelog $\langle \Delta^-(u), \Delta^+(u) \rangle$.

```
1.  $\Delta^-(u), \Delta^+(u) \leftarrow \emptyset$ ;
2.  $Relev(R) \leftarrow \{\Psi \in M \mid \Psi \text{ is pivot-relevant to } R \vee \Psi \text{ is relation-relevant to } R\}$ ;
3. foreach  $\Psi \in Relev(R)$  do
4.    $\Delta_{pivot}^-[\Psi](u), \Delta_{rel}^-[\Psi](u) \leftarrow \emptyset$ ;
5.    $\Delta_{pivot}^+[\Psi](u), \Delta_{rel}^+[\Psi](u) \leftarrow \emptyset$ ;
6.   // 1. Pivot-relevant contribution
7.   if  $\Psi$  is pivot-relevant to updates on  $R$  then
8.     foreach  $d \in D$  do
9.        $\Delta_{pivot}^-[\Psi](d) \leftarrow \{(s, p, o, g) \in W_0 \mid s = URI(d) \wedge g = URI(\Psi)\}$ ;
10.       $\Delta_{pivot}^-[\Psi](u) \leftarrow \Delta_{pivot}^-[\Psi](u) \cup \Delta_{pivot}^-[\Psi](d)$ ;
11.    end
12.    foreach  $i \in I$  do
13.       $\Delta_{pivot}^+[\Psi](i) \leftarrow \{(s, p, o, g) \mid (s, p, o) \in \Psi[r \leftarrow i](\sigma_1) \wedge g = URI(\Psi)\}$ ;
14.       $\Delta_{pivot}^+[\Psi](u) \leftarrow \Delta_{pivot}^+[\Psi](u) \cup \Delta_{pivot}^+[\Psi](i)$ ;
15.    end
16.  end
17.  // 2. Relation-relevant contribution
18.  if  $\Psi$  is relation-relevant to updates on  $R$  then
19.     $R^* \leftarrow pivot(\Psi)$ ;
20.    // 2.1 Affected pivot tuples for deletion
21.    if  $R$  occurs only once in  $path(\Psi)$  then
22.       $A_{\Psi}^- \leftarrow \{p^* \in R^*(\sigma_1) \mid \exists d \in D, \exists (s, p, o) \in \Psi[r^* \leftarrow p^*, r_R \leftarrow d](\sigma_1)\}$ ;
23.    end
24.    else
25.       $\sigma_R^0 \leftarrow (R(\sigma_1) \cup D) \setminus I$ ;
26.       $A_{\Psi}^- \leftarrow \{p^* \in R^*(\sigma_1) \mid \exists d \in D, \exists r_k \in Var_R(\Psi), \exists (s, p, o) \in \Psi[r^* \leftarrow p^*, r_k \leftarrow d](\sigma_1[R \leftarrow \sigma_R^0])\}$ ;
27.    end
28.    foreach  $p^* \in A_{\Psi}^-$  do
29.       $S_1[\Psi](p^*) \leftarrow \{(s, p, o, g) \in W_0 \mid s = URI(p^*) \wedge g = URI(\Psi)\}$ ;
30.       $S_2[\Psi](p^*) \leftarrow \{(s, p, o, g) \mid (s, p, o) \in \Psi[r^* \leftarrow p^*](\sigma_1) \wedge g = URI(\Psi)\}$ ;
31.       $\Delta_{rel}^-[\Psi](p^*) \leftarrow S_1[\Psi](p^*) \setminus S_2[\Psi](p^*)$ ;
32.       $\Delta_{rel}^-[\Psi](u) \leftarrow \Delta_{rel}^-[\Psi](u) \cup \Delta_{rel}^-[\Psi](p^*)$ ;
33.    end
34.    // 2.2 Affected pivot tuples for insertion
35.     $A_{\Psi}^+ \leftarrow \{p^* \in R^*(\sigma_1) \mid \exists i \in I, \exists r_k \in Var_R(\Psi), \exists (s, p, o) \in \Psi[r^* \leftarrow p^*, r_k \leftarrow i](\sigma_1)\}$ ;
36.    foreach  $p^* \in A_{\Psi}^+$  do
37.       $\Delta_{rel}^+[\Psi](p^*) \leftarrow \{(s, p, o, g) \mid (s, p, o) \in \Psi[r^* \leftarrow p^*](\sigma_1) \wedge g = URI(\Psi)\}$ ;
38.       $\Delta_{rel}^+[\Psi](u) \leftarrow \Delta_{rel}^+[\Psi](u) \cup \Delta_{rel}^+[\Psi](p^*)$ ;
39.    end
40.  end
41.  // 3. Overall deletion and insertion contributions
42.   $\Delta^-[\Psi](u) \leftarrow \Delta_{pivot}^-[\Psi](u) \cup \Delta_{rel}^-[\Psi](u)$ ;
43.   $\Delta^+[\Psi](u) \leftarrow \Delta_{pivot}^+[\Psi](u) \cup \Delta_{rel}^+[\Psi](u)$ ;
44.   $\Delta^-(u) \leftarrow \Delta^-(u) \cup \Delta^-[\Psi](u)$ ;
45.   $\Delta^+(u) \leftarrow \Delta^+(u) \cup \Delta^+[\Psi](u)$ ;
46. end
47. return  $\langle \Delta^-(u), \Delta^+(u) \rangle$ ;
```
